



VERIFYID: ONLINE BARANGAY SYSTEM WITH IDENTITY AUTHENTICATION THROUGH INTEGRATION OF IMAGE RECOGNITION

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PROJECT CONTEXT

Barangay authentication processes often rely on manual verification, which can be time-consuming and prone to errors. This study introduces VerifyID, an online barangay system integrating facial recognition for identity authentication and certificate processing to enhance efficiency and security.



OBJECTIVE OF THE STUDY

- Develop an online barangay system with facial recognition for identity authentication.
- Evaluate the system's performance based on ISO/IEC 25010 quality metrics.
- Identify the most effective image recognition model for accurate and fast verification.
- Assess user satisfaction and system usability through survey-based evaluation.



SCOPE AND LIMITATIONS

- VerifyID is an online barangay system for Barangay 781 Zone 85 residents.
- Integrates facial recognition (CNN for accuracy, YOLOv8 for fast detection) for identity authentication.
- Manages certificate requests, news dissemination, and inquiries to improve efficiency and security.
- Ensures accurate identity verification by matching uploaded IDs with live facial scans.
- Key features:
 - User registration with facial authentication
 - Certificate request management
 - Announcements and news updates
 - Admin dashboard
 - Inquiry handling
- System effectiveness is assessed through user testing and performance metrics to ensure accessibility, security, and efficiency.

- Exclusive to Barangay 781 Zone 85.
- Handling long names
- Facial recognition accuracy depends on:
 - Image quality (blurry or low-resolution images affect verification).
 - Lighting conditions (shadows or glare may cause recognition errors).
- Does not process financial transactions; payments are handled manually by barangay personnel.
- Limited to nine certification types; new document types require system updates.
- Manual approval of uploaded documents may cause delays.
- Only specific ID types accepted (PhilHealth, National ID, SSS, Student ID); black-and-white copies may not be recognized.

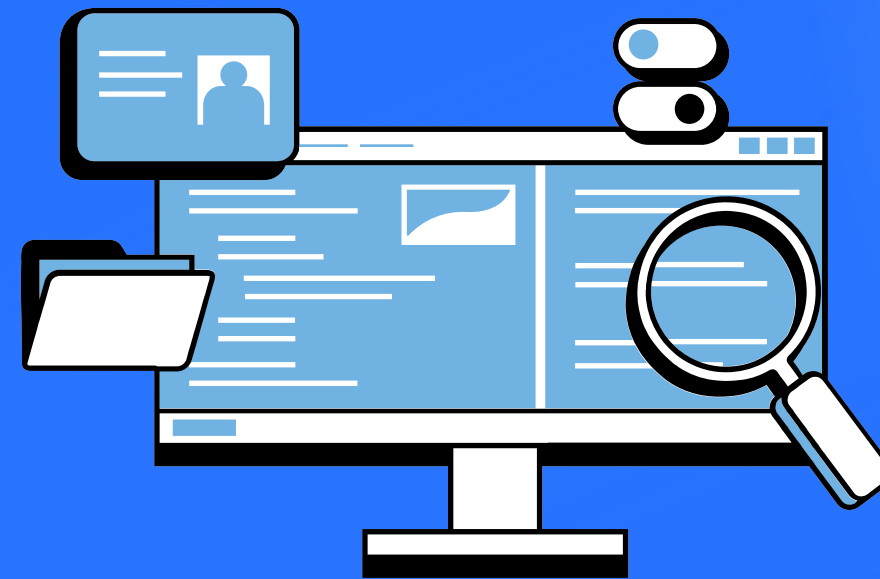




RESEARCH DESIGN



Descriptive
Research



Developmental
Research

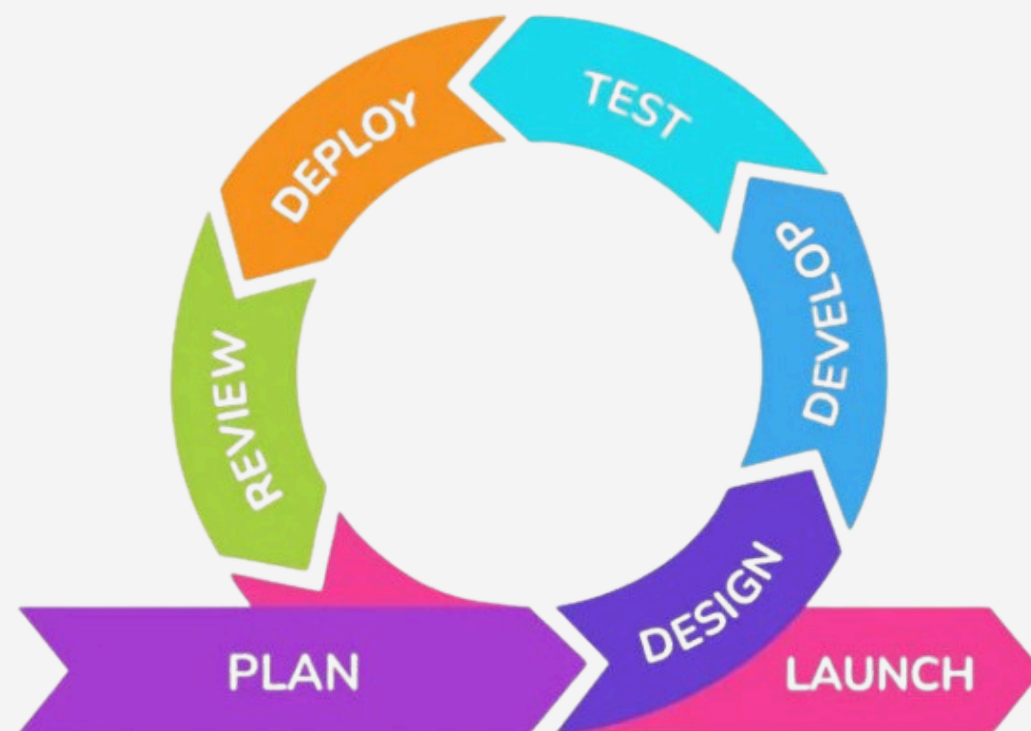


Experimental
Research





METHODOLOGY



Agile Methodology

PLAN

- The need for a secure and efficient barangay system was identified, leading to the definition of the project scope and objectives. Background research on digital governance and biometric authentication was conducted, along with gathering stakeholder input from barangay officials and residents.

DESIGN

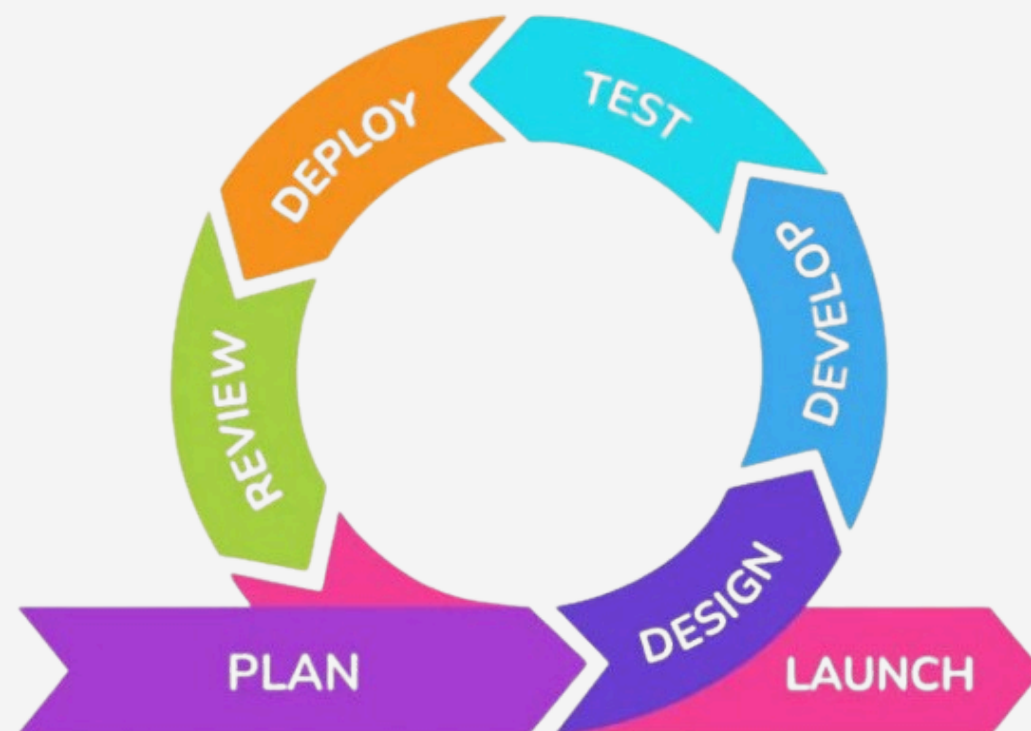
- The system architecture, database schema, and user interface for VerifyID were designed to ensure accessibility, usability, and compatibility across devices. Consistency in design elements was maintained, and system interactions were visualized through various diagrams.

DEVELOP

- Facial recognition using YOLOv8 and CNN was implemented for identity verification, with the model trained and fine-tuned for accuracy. Challenges such as image quality and false positives were addressed, ensuring seamless integration with the application for registration and service requests.



METHODOLOGY



Agile Methodology

TEST

- Functional and non-functional testing were performed to ensure system reliability, including unit, integration, performance, and usability testing. Tools like BlazeMeter, Loadster, Gatling, and BrowserStack were used to assess functionality, stability, and compatibility across devices.

DEPLOY

- The system architecture, database schema, and user interface for VerifyID were designed to ensure accessibility, usability, and compatibility across devices. Consistency in design elements was maintained, and system interactions were visualized through various diagrams.

REVIEW

- Progress was validated against business and functional requirements, ensuring quality, usability, and completeness of user stories. Sprint review meetings were held at the end of each sprint, where the development team presented system features, and barangay officials provided feedback to refine the product backlog.

LAUNCH

- This phase marks the transition of the system from development and testing to full operational deployment, making it accessible to end-users.



DATA COLLECTION AND ANALYTICAL TECHNIQUES



ANALYTICAL TECHNIQUES



DATA COLLECTION METHOD



Survey



Descriptive statistics



ANALYTICAL TECHNIQUES

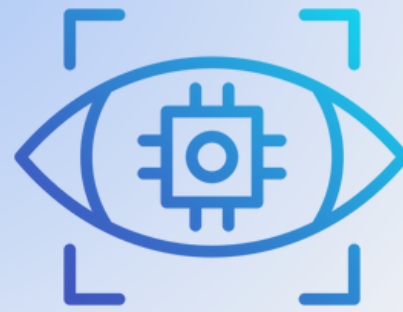


Confusion Matrix

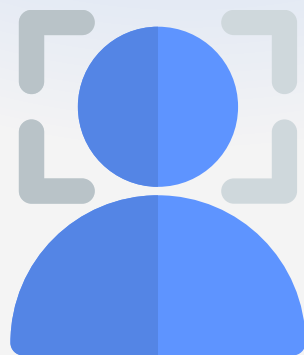




ALGORITHMS



CNN

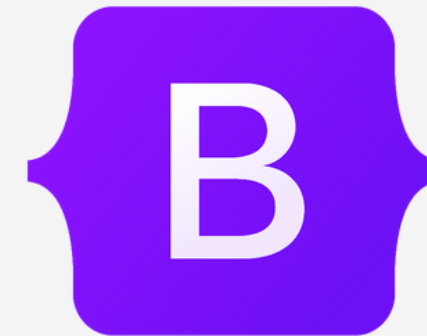


Faster R-CNN

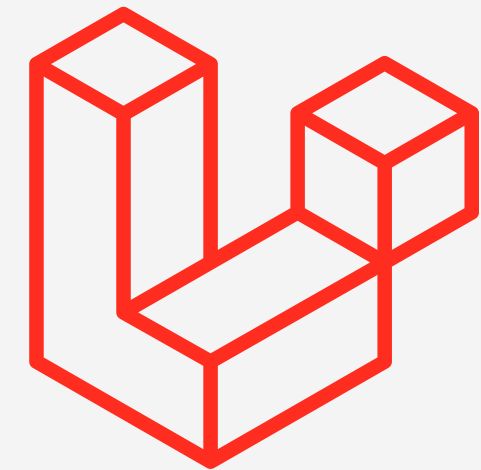


YOLOv8

FRAMEWORKS



Bootstrap v5.3



Laravel v11.0



DEMO



BARANGAY 781 ZONE 85 WEBSITE





RESULTS AND DISCUSSION

THE RESEARCHER DISCUSSES THE ACTIVITIES UNDERTAKEN TO FULFILL THE THREE SPECIFIC OBJECTIVES OF THIS STUDY. ALL OBJECTIVES ARE THOROUGHLY EXAMINED, WITH A FOCUS ON THE METHODOLOGIES, TOOLS, AND PRELIMINARY FINDINGS THAT WERE OBTAINED DURING THE EVALUATION PHASE.

OBJECTIVE 1

Examine and understand the Process of the Barangay system and Image Recognition.

OBJECTIVE 2

Explore existing systems or applications relevant to the proposed system.

OBJECTIVE 3

Analyze, understand, and perform preprocessing activities on the data gathered towards the development of the VerifyID.





RESULTS OF TRAINED ALGORITHM

CNN

Precision	1.000
Recall	1.000
F1 Score	1.000
Accuracy	1.000

YOLOv8

Precision	0.400
Recall	1.000
F1 Score	0.5714
Accuracy	0.400

Fast-R CNN

Precision	0.600
Recall	1.000
F1 Score	0.750
Accuracy	0.600



FUNCTIONAL

1. Unit Testing

- Facial recognition correctly extracted features and matched stored data.

2. Integration Testing

- Frontend, backend, and database integration worked smoothly.

3. System Testing

- Full registration and login workflow performed as expected.

4. Acceptance Testing

- Barangay residents successfully created accounts and requested certificates.

NON-FUNCTIONAL

1. Performance Testing

- System responded quickly under normal and peak loads.

2. Load Testing

- System handled high concurrent users without performance degradation.

3. Stress Testing

- System responded quickly under normal and peak loads.

4. Scalability Testing

- System adapted to increasing users and data volume without slowdowns.

5. Usability Testing

- System responded quickly under normal and peak loads.





MEAN RATING

METRICS	MEAN RATINGS	VERBAL INTERPRETATION
Functional Suitability	4.70	Strongly Agree
Performance Efficiency	4.64	Strongly Agree
Compatibility	4.60	Strongly Agree
Security	4.49	Strongly Agree
Average	4.61	Strongly Agree





CENTRAL TENDENCY

METRICS	MEAN	MEDIAN	MODE
Functional Suitability	4.70	4.75	4.76
Performance Efficiency	4.64	4.73	None
Compatibility	4.60	4.70	None
Security	4.49	4.70	4.15, 4.25
Overall	4.61	4.62	None





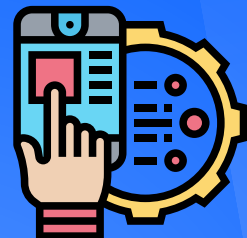
SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATION

Summary of Findings

- Functional testing confirmed that all system components like registration, certificate requests, and notifications worked as intended.
- Non-functional testing ensured that the system was responsive, scalable, and user-friendly across different devices and browsers.
- The system streamlines barangay operations by reducing manual tasks in certificate requests. 92.8% of users acknowledged its efficiency, while 92% reported smooth performance even with multiple concurrent users.
- The system was rated intuitive and easy to navigate by 92.2% of respondents. However, older residents and less tech-savvy users suggested adding guided walkthroughs, tooltips, and accessibility features.
- 92% of users confirmed system functionality across desktops, tablets, and smartphones. However, there was a demand for a dedicated mobile application, as many residents primarily use smartphones for barangay transactions.
- The integration of automated certificate processing and notifications improved barangay service delivery.



RECOMMENDATION



Improve UI & Navigation



Alternative Authentication



Live Scan Feedback



Time-Limited Face Scan



Enhance Usability on Mobile



Help & Support Feature



Clear Error Messages



Familiar Design Patterns

